Isabela Chaves de Almeida

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PROFESSIONAL SUMMARY

Software Engineer specializing in backend systems, distributed cloud infrastructure, and data platforms. Combines foundational systems programming in C and Linux with expertise in digital compliance and data privacy regulations (LGPD). Proven experience architecting scalable microservices, automated IaC deployments, and high-throughput data processing workflows across AWS, Kubernetes, and bare-metal homelab environments.

TECHNICAL SKILLS

- **Languages:** Go (Golang), Kotlin, Python, C, SQL
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda, Presto/Athena, Iceberg/s3 tables, Glue, kinesis), OpenTofu / Terraform, Docker, Kubernetes, Linux Systems Administration
- Databases & Storage: PostgreSQL (pgvector), Apache Iceberg, S3 Object Storage, Redis
- AI & Retrieval Engineering: RAG Systems, LangChain, Vector Embeddings, Cross-Encoder Rerankers
- **Observability & DevOps:** Prometheus, Grafana, Git/GitHub, CI/CD Automation
- **Architecture & Hardware:** Distributed Systems, RESTful APIs, Homelab Infrastructure, Digital Logic / FPGA (Tang Nano 20K)
- **Compliance & Governance:** Privacy by Design, LGPD Digital Evidence & Chain-of-Custody

WORK EXPERIENCE

Software Engineer (Part-time)

Neurodivertindo | Remote | Jan 2026 – Present

- Architected and maintained cloud backend services and analytical infrastructure using Go (Gin), Python, and AWS.
- Automated cloud provisioning using OpenTofu (IaC), deploying cost-effective analytical querying over S3 data lakes with Presto/Athena and Apache Iceberg.
- Designed and deployed an autoscaling, self-hosted CI/CD pipeline using OpenTofu, boto3, and custom EC2 AMIs to orchestrate Gitlab runners, significantly reducing build times and compute costs.
- Integrated system-wide metrics, tracing, and alerting with Prometheus and Grafana dashboards to monitor latency, token usage, and API availability.
- Built an interactive mind map explorer powered by vector embeddings and cosine similarity to render thousands of dynamic graph nodes and edges with sub-second retrieval latency.
- Architected and deployed an automated RAG (Retrieval-Augmented Generation) pipeline using LangChain, vector embeddings, pgvector, and cross-encoder rerankers orchestrated with serverless AWS Lambda functions.

Software Developer (Full-time)

Tribunal de Justiça do Estado de Alagoas | Remote | Jul 2026 – Present

- Engineered containerized backend services using Go deployed on Kubernetes clusters.
- Designed resilient REST APIs and microservice workflows aligned with strict data security, high availability, and compliance standards.
- Collaborated on distributed systems architecture to optimize database queries and service inter-communication.

Judicial Assistant / Legal Officer (Full-time)

Tribunal de Justiça do Estado do Ceará | Acopiara, CE | Oct 2023 – Nov 2025

- Architected an on-premise, privacy-compliant AI transcription and summarization pipeline using Python, Hugging Face Transformers, Whisper-v3-turbo, and a local Gemma LLM, reducing hearing review and decision-drafting turnaround by nearly 50%.
- Drafted judicial decisions, rulings, and legal documentation specializing in the intersection of law and technology.
- Evaluated compliance frameworks concerning LGPD (General Data Protection Law) in the criminal sector and chain-of-custody protocols for digital evidence.

FEATURED PROJECTS & SYSTEMS

Self-Hosted Bare-Metal Homelab & Data Platform

- Architected a self-hosted physical infrastructure running Kubernetes clusters and analytical data pipelines for stress-testing distributed services.
- Configured local persistent volumes, network topology, and automated monitoring using Prometheus and Grafana.

16-Bit Processor Architecture on Tang Nano 20K FPGA

- Implementing a custom 16-bit processor in hardware to explore instruction set architecture, datapath execution, and low-level digital logic design beneath the operating system layer.

EDUCATION

Bachelor of Laws (LL.B.)

Universidade Regional do Cariri (URCA) | Iguatu, CE | GPA: 9.2 / 10 | Mar 2021 – Sep 2023

CONTINUOUS LEARNING & COURSES

- Cloud & DevOps: AWS Solutions Architect Associate (Coursework), Kubernetes Administrator (CKA Coursework)
- Data Engineering & AI: AWS Data Engineer Associate (Coursework), Data Engineering 101, Machine Learning with Scikit-Learn